

The fontscale package

A flexible interface for setting font sizes

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1 Loading the package

```
\NeedsTeXFormat{LaTeX2e}[2023-11-01]
\ProvidesExplPackage{fontscale}{2025-09-07}{5.0.0}
  {A flexible interface for setting font sizes}

\msg_new:nnn { fontscale } { l3kernel-out-of-date }
{
  The~ fontscale~ package~ could~ not~ load. \\
  This~ package~ requires~
  L3~ programming~ layer~ version~ 2023-11-09~ or~ newer.
}

\msg_new:nnn { fontscale } { incompatible-package }
{ The~ fontscale~ package~ is~ incompatible~ with~ the~ '#1'~ package. }

\IfExplAtLeastTF { 2023-11-09 } { }
{ \msg_critical:nn { fontscale } { l3kernel-out-of-date } }
\hook_gput_code:nnn { package / scalefont / before } { fontscale }
{ \msg_error:nnn { fontscale } { incompatible-package } { scalefont } }
```

2 Messages

```
% Messages used in \fontscalesetup.
\msg_new:nnn { fontscale } { musical-base-invalid }
{
  Invalid~ value~ '#1'~ \msg_line_context:. \\
  The~ value~ of~ the~ key~ 'musical / base'~ must~ be~ a~ positive~ length~
  and~ less~ than~ 2048pt.
}

\msg_new:nnn { fontscale } { musical-ratio-invalid }
{
  Invalid~ value~ '#1'~ \msg_line_context:. \\
  The~ value~ of~ the~ key~ 'musical / ratio'~ must~ be~ greater~ than~ 1.
}

\msg_new:nnn { fontscale } { musical-notes-invalid }
{
  Invalid~ value~ '#1'~ \msg_line_context:. \\
  The~ value~ of~ the~ key~ 'musical / notes'~ must~ be~ a~ positive~
  integer.
}
```

```

    }
\msg_new:nnn { fontscale } { magscale-invalid }
{
  Invalid~ value~ '#1'~ \msg_line_context:. \\
  The~ value~ of~ the~ key~ 'magscale'~ must~ be~ positive.
}
\msg_new:nnn { fontscale } { name-font-size-invalid }
{
  Invalid~ font~ size~ of~ \iow_char:N \\#1~ '#2'~ \msg_line_context:. \\
  The~ font~ size~ must~ be~ a~ positive~ length~ and~ less~ than~ 2048pt.
}
\msg_new:nnn { fontscale } { size-normalsize-ratio-overwritten }
{
  The~ font~ size~ set~ by~ the~ key~ '#1 / size-normalsize-ratio'~
  has~ been~ overwritten~ by~ the~ key~ '#1 / size'~ \msg_line_context:.
}
\msg_new:nnn { fontscale } { baselineskip-size-ratio-overwritten }
{
  The~ font~ baselineskip~ set~ by~ the~ key~ '#1 / baselineskip-size-ratio'~
  has~ been~ overwritten~ by~ the~ key~ '#1 / baselineskip'~
  \msg_line_context:.
}
\msg_new:nnn { fontscale } { font-sizes-wrong-order }
{
  The~ font~ sizes~ are~ not~ in~ the~ correct~ order~ \msg_line_context:. \\
  The~ lengths~ of~ the~ font~ sizes~ should~ be~ ordered~ from~
  \iow_char:N \\tiny~ to~ \iow_char:N \\Huge.
}

% Messages used elsewhere.
\msg_new:nnn { fontscale } { math-mode-error }
{ '\iow_char:N \\#1'~ cannot~ be~ used~ in~ math~ mode~ \msg_line_context:. }
\msg_new:nnn { fontscale } { font-size-invalid }
{
  Invalid~ font~ size~ '#1'~ \msg_line_context:. \\
  The~ font~ size~ must~ be~ a~ positive~ length~ and~ less~ than~ 2048pt.
}
\msg_new:nnn { fontscale } { change-in-font-step-invalid }
{
  Invalid~ change~ in~ font~ step~ '#1'~ \msg_line_context:. \\
  The~ change~ in~ font~ step~ cannot~ be~
  greater~ than~ 9~ or~ less~ than~ -9.
}
\msg_new:nnn { fontscale } { current-font-step-invalid }
{
  The~ font~ size~ could~ not~ be~ stepped~ \msg_line_context:. \\
  The~ current~ font~ size~ '#1'~ does~ not~ equal~ the~ font~ size~ of~ any~
  font~ size~ command~ from~ \iow_char:N \\tiny~ to~ \iow_char:N \\Huge.
}

```

```

}
\msg_new:nnn { fontscale } { font-step-too-small }
{
  Using~ \iow_char:N \tiny~ \msg_line_context:. \
  The~ font~ size~ cannot~ be~ stepped~ to~ a~ size~ smaller~ than~
  \iow_char:N \tiny.
}
\msg_new:nnn { fontscale } { font-step-too-large }
{
  Using~ \iow_char:N \Huge~ \msg_line_context:. \
  The~ font~ size~ cannot~ be~ stepped~ to~ a~ size~ larger~ than~
  \iow_char:N \Huge.
}

```

3 Some variables

% Declares and initializes the font size and font baselineskip of each font
 % size command from \tiny to \Huge.

```

\dim_new:N \l_fontscale_tiny_size_dim
\dim_new:N \l_fontscale_scriptsize_size_dim
\dim_new:N \l_fontscale_footnotesize_size_dim
\dim_new:N \l_fontscale_small_size_dim
\dim_new:N \l_fontscale_normalsize_size_dim
\dim_new:N \l_fontscale_large_size_dim
\dim_new:N \l_fontscale_Large_size_dim
\dim_new:N \l_fontscale_LARGE_size_dim
\dim_new:N \l_fontscale_huge_size_dim
\dim_new:N \l_fontscale_Huge_size_dim

\skip_new:N \l_fontscale_tiny_baselineskip_skip
\skip_new:N \l_fontscale_scriptsize_baselineskip_skip
\skip_new:N \l_fontscale_footnotesize_baselineskip_skip
\skip_new:N \l_fontscale_small_baselineskip_skip
\skip_new:N \l_fontscale_normalsize_baselineskip_skip
\skip_new:N \l_fontscale_large_baselineskip_skip
\skip_new:N \l_fontscale_Large_baselineskip_skip
\skip_new:N \l_fontscale_LARGE_baselineskip_skip
\skip_new:N \l_fontscale_huge_baselineskip_skip
\skip_new:N \l_fontscale_Huge_baselineskip_skip

\dim_set:Nn \l_fontscale_tiny_size_dim { 6pt }
\dim_set:Nn \l_fontscale_scriptsize_size_dim { 7pt }
\dim_set:Nn \l_fontscale_footnotesize_size_dim { 8pt }
\dim_set:Nn \l_fontscale_small_size_dim { 9pt }
\dim_set:Nn \l_fontscale_normalsize_size_dim { 10pt }

```

```

\dim_set:Nn \l_fontscale_large_size_dim { 11pt }
\dim_set:Nn \l_fontscale_Large_size_dim { 12pt }
\dim_set:Nn \l_fontscale_LARGE_size_dim { 14pt }
\dim_set:Nn \l_fontscale_huge_size_dim { 16pt }
\dim_set:Nn \l_fontscale_Huge_size_dim { 18pt }

\skip_set:Nn \l_fontscale_tiny_baselineskip_skip { 7.2pt }
\skip_set:Nn \l_fontscale_scriptsize_baselineskip_skip { 8.4pt }
\skip_set:Nn \l_fontscale_footnotesize_baselineskip_skip { 9.6pt }
\skip_set:Nn \l_fontscale_small_baselineskip_skip { 10.8pt }
\skip_set:Nn \l_fontscale_normalsize_baselineskip_skip { 12 pt }
\skip_set:Nn \l_fontscale_large_baselineskip_skip { 13.2pt }
\skip_set:Nn \l_fontscale_Large_baselineskip_skip { 14.4pt }
\skip_set:Nn \l_fontscale_LARGE_baselineskip_skip { 16.8pt }
\skip_set:Nn \l_fontscale_huge_baselineskip_skip { 19.2pt }
\skip_set:Nn \l_fontscale_Huge_baselineskip_skip { 21.6pt }

% Stores the current font size and font baselineskip. They are set in the
% selectfont hook.
\dim_new:N \l_fontscale_size_dim
\skip_new:N \l_fontscale_baselineskip_skip
\hook_gput_code:nnn { selectfont } { fontscale }
{
  \dim_set:Nn \l_fontscale_size_dim { \f@size pt }
  \skip_set:Nn \l_fontscale_baselineskip_skip { \f@baselineskip }
}

% Used only to speed up floating point computations.
\fp_new:N \l__fontscale_musical_base_fp
\fp_new:N \l__fontscale_musical_notes_fp

% Scratch variable
\fp_new:N \l__fontscale_tmp_fp

```

4 Some functions

```

% Variants of l3kernel functions
\cs_generate_variant:Nn \msg_error:nnnn { nnnv }
% Similar to \fontsize + \selectfont, except that it:
% (1) Takes dimen and skip expressions as arguments without appending a default
% unit of 'pt'.
% (2) Avoids the issue where \f@size is set to the new font size before the
% second argument is expanded.
% (3) Issues a user-friendly error message if the font size is invalid.
\cs_new_protected:Npn \__fontscale_set_font_size:nn #1#2
{

```

```

\use:e
{
  \__fontscale_set_font_size_aux:nn { \dim_eval:n {#1} }
  { \skip_eval:n {#2} }
}
}
\cs_new_protected:Npn \__fontscale_set_font_size_aux:nn #1#2
{
  \dim_compare:nTF { \c_zero_dim < #1 < 2048pt }
  { \fontsize {#1} {#2} \selectfont }
  { \msg_error:nnn { fontscale } { font-size-invalid } {#1} }
}
% Similar to \__fontscale_set_font_size:nn, except that it sets the font
% baselineskip equal to the new font size times
% \l__fontscale_baselineskip_size_ratio_fp.
\cs_new_protected:Npn \__fontscale_set_font_size:n #1
{ \exp_args:Ne \__fontscale_set_font_size_aux:n { \dim_eval:n {#1} } }
\cs_new_protected:Npn \__fontscale_set_font_size_aux:n #1
{
  \dim_compare:nTF { \c_zero_dim < #1 < 2048pt }
  {
    \fontsize {#1}
    {
      \fp_to_dim:n
      { \dim_to_fp:n {#1} * \l__fontscale_baselineskip_size_ratio_fp }
    }
    \selectfont
  }
  { \msg_error:nnn { fontscale } { font-size-invalid } {#1} }
}
}

```

5 Define keys

```

\str_new:N \l__fontscale_typographic_scale_str
\str_new:N \l__fontscale_classic_base_str
\str_new:N \l__fontscale_classic_point_str

\keys_define:nn { fontscale }
{
  magscale .tl_set:N = \l__fontscale_magscale_tl
  , magscale .value_required:n = true

  , baselineskip-size-ratio .fp_set:N =
    \l__fontscale_baselineskip_size_ratio_fp
  , baselineskip-size-ratio .value_required:n = true
}

```

```

, typographic-scale .choices:nn = { classic , musical }
  { \str_set:Nn \l__fontscale_typographic_scale_str {#1} }
, typographic-scale .value_required:n = true

, classic .meta:n = { typographic-scale = classic }
, classic .value_forbidden:n = true

, musical .meta:n = { typographic-scale = musical }
, musical .value_forbidden:n = true
}
\keys_define:nn { fontscale / classic }
{
  base .choices:nn = { 10 , 11 , 12 }
    { \str_set:Nn \l__fontscale_classic_base_str {#1} }
, base .value_required:n = true

, point .choices:nn = { pt , bp , dd , nd }
    { \str_set:Nn \l__fontscale_classic_point_str {#1} }
, point .value_required:n = true
}
\keys_define:nn { fontscale / musical }
{
  base .dim_set:N = \l__fontscale_musical_base_dim
, base .value_required:n = true

, ratio .fp_set:N = \l__fontscale_musical_ratio_fp
, ratio .value_required:n = true

, notes .int_set:N = \l__fontscale_musical_notes_int
, notes .value_required:n = true
}
\keys_define:nn { fontscale / normalsize }
{
  size .tl_set:N = \l__fontscale_normalsize_size_tl
, size .value_required:n = true

, baselineskip-size-ratio .tl_set:N =
  \l__fontscale_normalsize_baselineskip_size_ratio_tl
, baselineskip-size-ratio .value_required:n = true

, baselineskip .tl_set:N = \l__fontscale_normalsize_baselineskip_tl
, baselineskip .value_required:n = true
}
\tl_map_inline:nn
{
  {tiny} {scriptsize} {footnotesize} {small}

```

```

{large} {Large} {LARGE} {huge} {Huge}
}
{
\keys_define:nn { fontscale / #1 }
{
    size-normalsize-ratio .tl_set:c =
        { l__fontscale_#1_size_normalsize_ratio_tl }
    , size-normalsize-ratio .value_required:n = true

    , size .tl_set:c = { l__fontscale_#1_size_tl }
    , size .value_required:n = true

    , baselineskip-size-ratio .tl_set:c =
        { l__fontscale_#1_baselineskip_size_ratio_tl }
    , baselineskip-size-ratio .value_required:n = true

    , baselineskip .tl_set:c = { l__fontscale_#1_baselineskip_tl }
    , baselineskip .value_required:n = true
}
}
\keys_define:nn { fontscale }
{
    tiny .tl_set:N = \l__fontscale_tiny_size_tl
    , tiny .value_required:n = true

    , scriptsize .tl_set:N = \l__fontscale_scriptsize_size_tl
    , scriptsize .value_required:n = true

    , footnotesize .tl_set:N = \l__fontscale_footnotesize_size_tl
    , footnotesize .value_required:n = true

    , small .tl_set:N = \l__fontscale_small_size_tl
    , small .value_required:n = true

    , normalsize .tl_set:N = \l__fontscale_normalsize_size_tl
    , normalsize .value_required:n = true

    , large .tl_set:N = \l__fontscale_large_size_tl
    , large .value_required:n = true

    , Large .tl_set:N = \l__fontscale_Large_size_tl
    , Large .value_required:n = true

    , LARGE .tl_set:N = \l__fontscale_LARGE_size_tl
    , LARGE .value_required:n = true
}

```

```

, huge .tl_set:N = \l__fontscale_huge_size_tl
, huge .value_required:n = true

, Huge .tl_set:N = \l__fontscale_Huge_size_tl
, Huge .value_required:n = true
}

% Pre-compiles the keys with their initial values and then sets the keys to
% their initial values. This is significantly faster than using .initial:n and
% \keys_precompile:nnN. The special value of \q_no_value is used to test if the
% key has been set by the user.
\cs_new_protected:Npn \__fontscale_keys_set_initial:
{
  \str_set:Nn \l__fontscale_typographic_scale_str { classic }
  \str_set:Nn \l__fontscale_classic_base_str { 10 }
  \str_set:Nn \l__fontscale_classic_point_str { pt }
  \dim_set:Nn \l__fontscale_musical_base_dim { 10pt }
  \fp_set:Nn \l__fontscale_musical_ratio_fp { 2 }
  \int_set:Nn \l__fontscale_musical_notes_int { 5 }
  \fp_set:Nn \l__fontscale_baselineskip_size_ratio_fp { 1.2 }

  \tl_map_inline:nn
  {
    \l__fontscale_tiny_size_normalsize_ratio_tl
    \l__fontscale_scriptsize_size_normalsize_ratio_tl
    \l__fontscale_footnotesize_size_normalsize_ratio_tl
    \l__fontscale_small_size_normalsize_ratio_tl
    \l__fontscale_large_size_normalsize_ratio_tl
    \l__fontscale_Large_size_normalsize_ratio_tl
    \l__fontscale_LARGE_size_normalsize_ratio_tl
    \l__fontscale_huge_size_normalsize_ratio_tl
    \l__fontscale_Huge_size_normalsize_ratio_tl

    \l__fontscale_tiny_size_tl
    \l__fontscale_scriptsize_size_tl
    \l__fontscale_footnotesize_size_tl
    \l__fontscale_small_size_tl
    \l__fontscale_normalsize_size_tl
    \l__fontscale_large_size_tl
    \l__fontscale_Large_size_tl
    \l__fontscale_LARGE_size_tl
    \l__fontscale_huge_size_tl
    \l__fontscale_Huge_size_tl

    \l__fontscale_tiny_baselineskip_size_ratio_tl
    \l__fontscale_scriptsize_baselineskip_size_ratio_tl
    \l__fontscale_footnotesize_baselineskip_size_ratio_tl

```



```

\l__fontscale_small_baselineskip_size_ratio_tl
\l__fontscale_normalsize_baselineskip_size_ratio_tl
\l__fontscale_large_baselineskip_size_ratio_tl
\l__fontscale_Large_baselineskip_size_ratio_tl
\l__fontscale_LARGE_baselineskip_size_ratio_tl
\l__fontscale_huge_baselineskip_size_ratio_tl
\l__fontscale_Huge_baselineskip_size_ratio_tl

\l__fontscale_tiny_baselineskip_tl
\l__fontscale_scriptsize_baselineskip_tl
\l__fontscale_footnotesize_baselineskip_tl
\l__fontscale_small_baselineskip_tl
\l__fontscale_normalsize_baselineskip_tl
\l__fontscale_large_baselineskip_tl
\l__fontscale_Large_baselineskip_tl
\l__fontscale_LARGE_baselineskip_tl
\l__fontscale_huge_baselineskip_tl
\l__fontscale_Huge_baselineskip_tl

\l__fontscale_magscale_tl
}
{ \tl_set:Nn ##1 { \q_no_value } }
}
\__fontscale_keys_set_initial:

```

6 Set and process keys

```

\NewDocumentCommand \fontscalesetup { s m }
{
  \mode_if_math:TF
  { \msg_error:nnn { fontscale } { math-mode-error } { fontscalesetup } }
  {
    \IfBooleanTF #1
    { \__fontscale_keys_reset:n {#2} }
    { \__fontscale_keys_set:n {#2} }
  }
}

\cs_new_protected:Npn \__fontscale_keys_reset:n
{ \__fontscale_keys_set_initial: \__fontscale_keys_set:n }

\cs_new_protected:Npn \__fontscale_keys_set:n #1
{
  \keys_set:nn { fontscale } {#1}
  \__fontscale_keys_process:
  \normalsize
}

\cs_new_protected:Npn \__fontscale_keys_process:
{

```

```

    \__fontscale_keys_process_start:
    \__fontscale_keys_process_sizes_normalsize:
    \__fontscale_keys_process_sizes_other:
    \__fontscale_keys_process_baselineskips:
    \__fontscale_keys_process_end:
  }
\cs_new:Npn \__fontscale_keys_process_break: #1 \normalsize { }
\cs_new_protected:Npn \__fontscale_keys_process_start:
{
  \dim_compare:nF { \c_zero_dim < \l__fontscale_musical_base_dim < 2048pt }
  {
    \msg_error:nnV { fontscale } { musical-base-invalid }
    \l__fontscale_musical_base_dim
    \__fontscale_keys_process_break:
  }
  \fp_compare:nNnF \l__fontscale_musical_ratio_fp > \c_one_fp
  {
    \msg_error:nne { fontscale } { musical-ratio-invalid }
    { \fp_to_tl:N \l__fontscale_musical_ratio_fp }
    \__fontscale_keys_process_break:
  }
  \int_compare:nNnF \l__fontscale_musical_notes_int > 0
  {
    \msg_error:nnV { fontscale } { musical-notes-invalid }
    \l__fontscale_musical_notes_int
    \__fontscale_keys_process_break:
  }
  \str_if_eq:VnT \l__fontscale_typographic_scale_str { musical }
  {
    \fp_set:Nn \l__fontscale_musical_base_fp
      { \dim_to_fp:n { \l__fontscale_musical_base_dim } }
    \fp_set:Nn \l__fontscale_musical_notes_fp
      { \int_use:N \l__fontscale_musical_notes_int }
  }
  \quark_if_no_value:NF \l__fontscale_magscale_tl
  {
    \fp_set:Nn \l__fontscale_tmp_fp { \l__fontscale_magscale_tl }
    \tl_set:NV \l__fontscale_magscale_tl \l__fontscale_tmp_fp
    \fp_compare:nNnF \l__fontscale_tmp_fp > \c_zero_fp
    {
      \msg_error:nne { fontscale } { magscale-invalid }
      { \fp_to_tl:N \l__fontscale_tmp_fp }
      \__fontscale_keys_process_break:
    }
  }
}
\cs_new_protected:Npn \__fontscale_keys_process_sizes_normalsize:
{
  \quark_if_no_value:NTF \l__fontscale_normalsize_size_tl

```

```

{
  \dim_set:Nn \l_fontscale_normalsize_size_dim
  {
    \str_case:Vn \l__fontscale_typographic_scale_str
    {
      { classic }
      {
        \l__fontscale_classic_base_str
        \l__fontscale_classic_point_str
      }
      { musical } { \l__fontscale_musical_base_dim }
    }
  }
}
{
  \dim_set:Nn \l_fontscale_normalsize_size_dim
  { \l__fontscale_normalsize_size_tl }
  \tl_set:NV \l__fontscale_normalsize_size_tl
  \l_fontscale_normalsize_size_dim
}
\quark_if_no_value:NF \l__fontscale_magscale_tl
{
  \dim_set:Nn \l_fontscale_normalsize_size_dim
  {
    \fp_to_dim:n
    {
      \dim_to_fp:n { \l_fontscale_normalsize_size_dim } *
      \l__fontscale_magscale_tl
    }
  }
}
\dim_compare:nF { \c_zero_dim < \l_fontscale_normalsize_size_dim < 2048pt }
{
  \msg_error:nnnV { fontscale } { name-font-size-invalid } { normalsize }
  \l_fontscale_normalsize_size_dim
  \__fontscale_keys_process_break:
}
}
\cs_new_protected:Npn \__fontscale_keys_process_sizes_other:
{
  \cs_set_eq:Nc \__fontscale_keys_process_sizes_other_aux:n
  {
    __fontscale_keys_process_sizes_other_
    \str_case:Vn \l__fontscale_typographic_scale_str
    {
      { classic } { classic_ \l__fontscale_classic_base_str :n }
      { musical } { musical:n }
    }
  }
}

```

```

\tl_map_function:nN
{
  {tiny} {scriptsize} {footnotesize} {small}
  {large} {Large} {LARGE} {huge} {Huge}
}
\__fontscale_keys_process_sizes_other:n
}
\cs_new_protected:cpn { __fontscale_keys_process_sizes_other_classic_10:n } #1
{
  \dim_set:cn { l_fontscale_#1_size_dim }
  {
    \str_case:nn {#1}
    {
      { tiny          } { 6 }
      { scriptsize    } { 7 }
      { footnotesize   } { 8 }
      { small          } { 9 }
      { large          } { 11 }
      { Large          } { 12 }
      { LARGE          } { 14 }
      { huge           } { 16 }
      { Huge           } { 18 }
    }
    \l__fontscale_classic_point_str
  }
}
\cs_new_protected:cpn { __fontscale_keys_process_sizes_other_classic_11:n } #1
{
  \dim_set:cn { l_fontscale_#1_size_dim }
  {
    \str_case:nn {#1}
    {
      { tiny          } { 7 }
      { scriptsize    } { 8 }
      { footnotesize   } { 9 }
      { small          } { 10 }
      { large          } { 12 }
      { Large          } { 14 }
      { LARGE          } { 16 }
      { huge           } { 18 }
      { Huge           } { 21 }
    }
    \l__fontscale_classic_point_str
  }
}
\cs_new_protected:cpn { __fontscale_keys_process_sizes_other_classic_12:n } #1
{
  \dim_set:cn { l_fontscale_#1_size_dim }
  {

```

```

\str_case:nn {#1}
{
  { tiny          } { 8 }
  { scriptsize    } { 9 }
  { footnotesize  } { 10 }
  { small         } { 11 }
  { large         } { 14 }
  { Large         } { 16 }
  { LARGE         } { 18 }
  { huge          } { 21 }
  { Huge          } { 24 }
}

\l__fontscale_classic_point_str
}
}

\cs_new_protected:Npn \__fontscale_keys_process_sizes_other_musical:n #1
{
  \dim_set:cn { l_fontscale_#1_size_dim }
  {
    \fp_to_dim:n
    {
      \l__fontscale_musical_base_fp * \l__fontscale_musical_ratio_fp ^
      (
        \str_case:nn {#1}
        {
          { tiny          } { -4 }
          { scriptsize    } { -3 }
          { footnotesize  } { -2 }
          { small         } { -1 }
          { large         } { 1 }
          { Large         } { 2 }
          { LARGE         } { 3 }
          { huge          } { 4 }
          { Huge          } { 5 }
        }
        / \l__fontscale_musical_notes_fp
      )
    }
  }
}

\cs_new_protected:Npn \__fontscale_keys_process_sizes_other:n #1
{
  \quark_if_no_value:cF { l_fontscale_#1_size_normalsize_ratio_tl }
  {
    \fp_set:Nn \l__fontscale_tmp_fp
    { \use:c { l_fontscale_#1_size_normalsize_ratio_tl } }
    \tl_set:cV { l_fontscale_#1_size_normalsize_ratio_tl }
    \l__fontscale_tmp_fp
    \dim_set:cn { l_fontscale_#1_size_dim }
  }
}

```

```

    {
      \fp_to_dim:n
      {
        \dim_to_fp:n { \l_fontscale_normalsize_size_dim } *
        \l__fontscale_tmp_fp
      }
    }
    \quark_if_no_value:cT { l__fontscale_#1_size_tl }
    { \prg_break:n { \use_none:n } }
  }
  \quark_if_no_value:cF { l__fontscale_#1_size_tl }
  {
    \dim_set:cn { l_fontscale_#1_size_dim }
    { \use:c { l__fontscale_#1_size_tl } }
    \tl_set:cv { l__fontscale_#1_size_tl } { l_fontscale_#1_size_dim }
    \quark_if_no_value:cF { l__fontscale_#1_size_normalsize_ratio_tl }
    {
      \msg_warning:nnn { fontscale }
      { size-normalsize-ratio-overwritten } {#1}
    }
    \prg_break:n { \use:n }
  }
  \__fontscale_keys_process_sizes_other_aux:n {#1}
  \use_ii:nn
  \prg_break_point:
  {
    \quark_if_no_value:NF \l__fontscale_magscale_tl
    {
      \dim_set:cn { l_fontscale_#1_size_dim }
      {
        \fp_to_dim:n
        {
          \dim_to_fp:n { \use:c { l_fontscale_#1_size_dim } } *
          \l__fontscale_magscale_tl
        }
      }
    }
  }
}
\dim_compare:nF
{ \c_zero_dim < \use:c { l_fontscale_#1_size_dim } < 2048pt }
{
  \tl_map_break:n
  {
    \msg_error:nnnv { fontscale } { name-font-size-invalid } {#1}
    { l_fontscale_#1_size_dim }
    \__fontscale_keys_process_break:
  }
}
}

```

```

\cs_new_protected:Npn \__fontscale_keys_process_baselineskips:
{
  \tl_map_function:nN
  {
    {tiny} {scriptsize} {footnotesize} {small} {normalsize}
    {large} {Large} {LARGE} {huge} {Huge}
  }
  \__fontscale_keys_process_baselineskips:n
}
\cs_new_protected:Npn \__fontscale_keys_process_baselineskips:n #1
{
  \quark_if_no_value:cF { l__fontscale_#1_baselineskip_size_ratio_tl }
  {
    \fp_set:Nn \l__fontscale_tmp_fp
    { \use:c { l__fontscale_#1_baselineskip_size_ratio_tl } }
    \tl_set:cv { l__fontscale_#1_baselineskip_size_ratio_tl }
    \l__fontscale_tmp_fp
    \skip_set:cn { l_fontscale_#1_baselineskip_skip }
    {
      \fp_to_dim:n
      {
        \dim_to_fp:n { \use:c { l_fontscale_#1_size_dim } } *
        \l__fontscale_tmp_fp
      }
    }
    \quark_if_no_value:cT { l__fontscale_#1_baselineskip_tl }
    { \prg_break: }
  }
  \quark_if_no_value:cF { l__fontscale_#1_baselineskip_tl }
  {
    \skip_set:cn { l_fontscale_#1_baselineskip_skip }
    { \use:c { l__fontscale_#1_baselineskip_tl } }
    \tl_set:cv { l__fontscale_#1_baselineskip_tl }
    { l_fontscale_#1_baselineskip_skip }
    \quark_if_no_value:cF { l__fontscale_#1_baselineskip_size_ratio_tl }
    {
      \msg_warning:nnn { fontscale }
      { baselineskip-size-ratio-overwritten } {#1}
    }
    \quark_if_no_value:NF \l__fontscale_magscale_tl
    {
      \skip_set:cn { l_fontscale_#1_baselineskip_skip }
      {
        \fp_to_dim:n
        {
          \dim_to_fp:n
          { \use:c { l_fontscale_#1_baselineskip_skip } }
          * \l__fontscale_magscale_tl
        }
      }
    }
  }
}

```

```

    }
  }
  \prg_break:
}
\skip_set:cn { l_fontscale_#1_baselineskip_skip }
{
  \fp_to_dim:n
  {
    \dim_to_fp:n { \use:c { l_fontscale_#1_size_dim } } *
    \l__fontscale_baselineskip_size_ratio_fp
  }
}
\prg_break_point:
}
\cs_new_protected:Npn \__fontscale_keys_process_end:
{
  \dim_compare:nF
  {
    \l_fontscale_tiny_size_dim
    < \l_fontscale_scriptsize_size_dim
    < \l_fontscale_footnotesize_size_dim
    < \l_fontscale_small_size_dim
    < \l_fontscale_normalsize_size_dim
    < \l_fontscale_large_size_dim
    < \l_fontscale_Large_size_dim
    < \l_fontscale_LARGE_size_dim
    < \l_fontscale_huge_size_dim
    < \l_fontscale_Huge_size_dim
  }
  { \msg_warning:nn { fontscale } { font-sizes-wrong-order } }
}

```

7 Document commands

% Defines the font size commands from \tiny to \Huge and initializes to
 % \normalsize. Sets \@currsz only for compatibility. The internal functions
 % are not used elsewhere for compatibility with hooks such
 % as \AddToHook{cmd//after}{<code>}.

```

\tl_map_inline:nn
{
  {tiny} {scriptsize} {footnotesize} {small} {normalsize}
  {large} {Large} {LARGE} {huge} {Huge}
}
{
  \use:e
  {
    \DeclareDocumentCommand \exp_not:c {#1} { }
  }
}

```



```

        {
            \exp_not:N \mode_if_math:TF
            { \msg_error:nnn { fontsize } { math-mode-error } {#1} }
            { \exp_not:c { __fontsize_#1: } }
        }
    }
\cs_new_protected:cpe { __fontsize_#1: }
{
    \cs_set_eq:NN \exp_not:N \@currsize \exp_not:c {#1}
    \exp_not:N \fontsize
    { \exp_not:N \dim_use:N \exp_not:c { l_fontscale_#1_size_dim } }
    {
        \exp_not:N \skip_use:N
        \exp_not:c { l_fontscale_#1_baselineskip_skip }
    }
    \exp_not:N \selectfont
}
}
\normalsize

\cs_new_eq:NN \currentfontsize \l_fontscale_size_dim
\cs_new_eq:NN \currentfontbaselineskip \l_fontscale_baselineskip_skip
\cs_new_eq:NN \currentnormalsize \l_fontscale_normalsize_size_dim

\NewDocumentCommand \setfontsize { o m }
{
    \mode_if_math:TF
    { \msg_error:nnn { fontsize } { math-mode-error } { setfontsize } }
    {
        \IfNoValueTF {#1}
        { \__fontsize_set_font_size:n {#2} }
        { \__fontsize_set_font_size:nn {#2} {#1} }
    }
}

\NewDocumentCommand \scalefont { m }
{
    \mode_if_math:TF
    { \msg_error:nnn { fontsize } { math-mode-error } { scalefont } }
    { \__fontsize_scale_font:n {#1} }
}
\cs_new_protected:Npn \__fontsize_scale_font:n #1
{
    \fp_set:Nn \l__fontsize_tmp_fp {#1}
    \__fontsize_set_font_size:nn
    {
        \fp_to_dim:n
        { \l__fontsize_tmp_fp * \dim_to_fp:n { \l_fontscale_size_dim } }
    }
}

```

```

    }
    {
      \fp_to_dim:n
      {
        \l__fontscale_tmp_fp *
        \dim_to_fp:n { \l_fontscale_baselineskip_skip }
      }
    }
  }
}

\NewDocumentCommand \stepfontsize { m }
{
  \mode_if_math:TF
  { \msg_error:nnn { fontscale } { math-mode-error } { stepfontsize } }
  { \__fontscale_step_font_size:n {#1} }
}

\cs_new_protected:Npn \__fontscale_step_font_size:n #1
{
  \use:e
  {
    \__fontscale_step_font_size:nn
    {
      \dim_case:nn { \l_fontscale_size_dim }
      {
        { \l_fontscale_tiny_size_dim } { -4 }
        { \l_fontscale_scriptsize_size_dim } { -3 }
        { \l_fontscale_footnotesize_size_dim } { -2 }
        { \l_fontscale_small_size_dim } { -1 }
        { \l_fontscale_normalsize_size_dim } { 0 }
        { \l_fontscale_large_size_dim } { 1 }
        { \l_fontscale_Large_size_dim } { 2 }
        { \l_fontscale_LARGE_size_dim } { 3 }
        { \l_fontscale_huge_size_dim } { 4 }
        { \l_fontscale_Huge_size_dim } { 5 }
      }
    }
    { \int_eval:n {#1} }
  }
}

\cs_new_protected:Npn \__fontscale_step_font_size:nn #1#2
{
  \bool_case:nF
  {
    { \tl_if_empty_p:n {#1} }
    {
      \msg_warning:nnV { fontscale } { current-font-step-invalid }
      \l_fontscale_size_dim
    }
  }
}

```

```

{ \int_compare_p:nNn { \int_abs:n {#2} } > 9 }
{ \msg_error:nnn { fontscale } { change-in-font-step-invalid } {#2} }
{ \int_compare_p:nNn { #1 + #2 } < { -4 } }
{
  \msg_warning:nn { fontscale } { font-step-too-small }
  \tiny
}
{ \int_compare_p:nNn { #1 + #2 } > 5 }
{
  \msg_warning:nn { fontscale } { font-step-too-large }
  \Huge
}
}
{
  \int_case:nn { #1 + #2 }
  {
    { -4 } { \tiny }
    { -3 } { \scriptsize }
    { -2 } { \footnotesize }
    { -1 } { \small }
    { 0 } { \normalsize }
    { 1 } { \large }
    { 2 } { \Large }
    { 3 } { \LARGE }
    { 4 } { \huge }
    { 5 } { \Huge }
  }
}
}

\NewDocumentCommand \setfontbaselineskip { m }
{
  \mode_if_math:TF
  {
    \msg_error:nnn { fontscale } { math-mode-error }
    { setfontbaselineskip }
  }
  { \_fontscale_set_font_baselineskip:n {#1} }
}

\cs_new_protected:Npn \_fontscale_set_font_baselineskip:n
{ \_fontscale_set_font_size:nn { \l_fontscale_size_dim } }

\NewDocumentCommand \printsampletext { s +m }
{
  \mode_if_math:TF
  { \msg_error:nnn { fontscale } { math-mode-error } { printsampletext } }
  {
    \IfBooleanTF #1

```

```

        { \_fontscale_print_sample_text_descending_order:n {#2} }
        { \_fontscale_print_sample_text_ascending_order:n {#2} }
    }
}
\cs_new_protected:Npn \_fontscale_print_sample_text_ascending_order:n #1
{
    \group_begin:
    \tl_map_inline:nn
    {
        \tiny \scriptsize \footnotesize \small \normalsize
        \large \Large \LARGE \huge \Huge
    }
    { ##1 #1 \par }
    \group_end:
}
\cs_new_protected:Npn \_fontscale_print_sample_text_descending_order:n #1
{
    \group_begin:
    \tl_map_inline:nn
    {
        \Huge \huge \LARGE \Large \large
        \normalsize \small \footnotesize \scriptsize \tiny
    }
    { ##1 #1 \par }
    \group_end:
}

\NewDocumentCommand \printfontsizecommand { }
{
    \mode_if_math:TF
    {
        \msg_error:nnn { fontscale } { math-mode-error }
        { printfontsizecommand }
    }
    { \_fontscale_print_name: }
}
\cs_new_protected:Npn \_fontscale_print_name:
{
    \c_backslash_str
    \dim_case:nnF { \l_fontscale_size_dim }
    {
        { \l_fontscale_tiny_size_dim } { { tiny } }
        { \l_fontscale_scriptsize_size_dim } { { scriptsize } }
        { \l_fontscale_footnotesize_size_dim } { { footnotesize } }
        { \l_fontscale_small_size_dim } { { small } }
        { \l_fontscale_normalsize_size_dim } { { normalsize } }
        { \l_fontscale_large_size_dim } { { large } }
        { \l_fontscale_Large_size_dim } { { Large } }
    }
}

```

```

    { \l_fontscale_LARGE_size_dim      } { LARGE \@      }
    { \l_fontscale_huge_size_dim       } { huge          }
    { \l_fontscale_Huge_size_dim       } { Huge          }
  }
  { undefined }
}

```

8 Purifying text

```

\NewExpandableDocumentCommand \_fontscale_use_none_sm:w { s m } { }
\NewExpandableDocumentCommand \_fontscale_use_none_om:w { o m } { }

\text_declare_purify_equivalent:Nn \fontscalesetup
  { \_fontscale_use_none_sm:w }
\text_declare_purify_equivalent:Nn \setfontsize { \_fontscale_use_none_om:w }
\text_declare_purify_equivalent:Nn \scalefont { \use_none:n }
\text_declare_purify_equivalent:Nn \stepfontsize { \use_none:n }
\text_declare_purify_equivalent:Nn \setfontbaselineskip { \use_none:n }

```